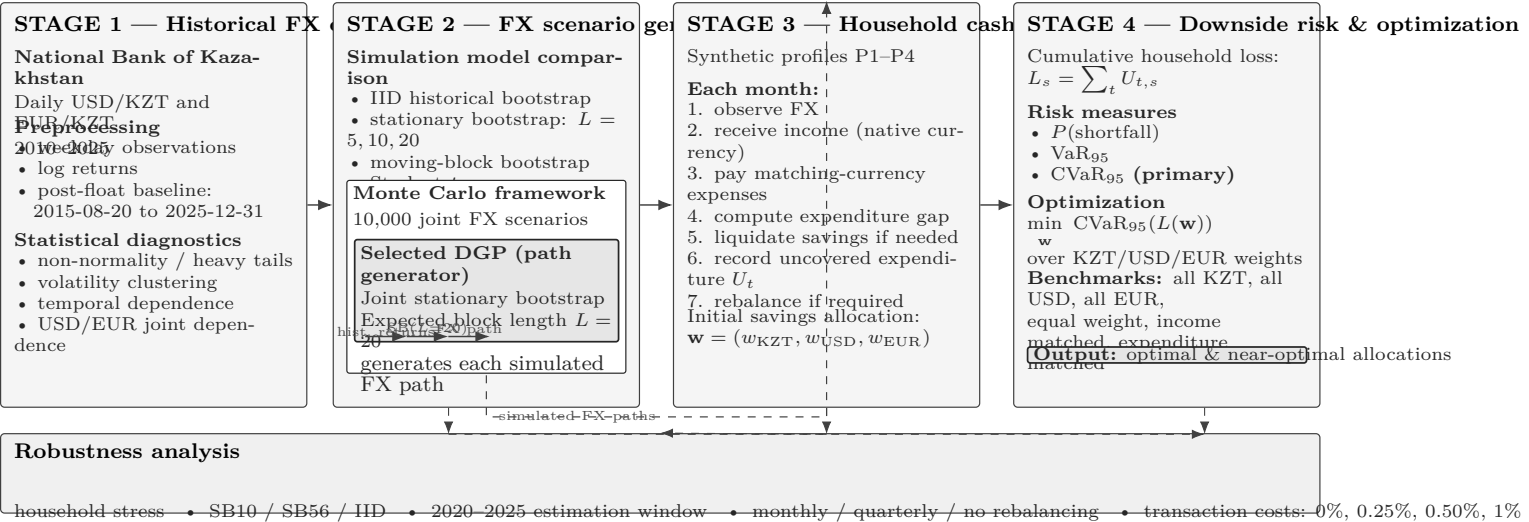
STAGE 1 — Historical FX
National Bank of Kazakhstan
Daily USD/KZT and EUR/KZT
Preprocessing
2010–2025
• weekday observations
• log returns
• post-float baseline: 2015-08-20 to 2025-12-31
Statistical diagnostics
• non-normality / heavy tails
• volatility clustering
• temporal dependence
• USD/EUR joint dependence
STAGE 2 — FX scenario ge
Simulation model comparison
• IID historical bootstrap
• stationary bootstrap: $L = 5, 10, 20$
• moving-block bootstrap
Monte Carlo framework
10,000 joint FX scenarios
Selected DGP (path generator)
Joint stationary bootstrap
Expected block length $L = 20$
hist. returns → 20/path
generates each simulated FX path
STAGE 3 — Household cash
Synthetic profiles P1–P4
Each month:
1. observe FX
2. receive income (native currency)
3. pay matching-currency expenses
4. compute expenditure gap
5. liquidate savings if needed
6. record uncovered expenditure U_t
7. rebalance if required
Initial savings allocation:
$\mathbf{w} = (w_{\mathrm{KZT}}, w_{\mathrm{USD}}, w_{\mathrm{EUR}})$
STAGE 4 — Downside risk & optimization
Cumulative household loss:
$L_s = \sum_t U_{t,s}$
Risk measures
• P(shortfall)
• VaR_{95}
• CVaR_{95} (primary)
Optimization
$\min_{\mathbf{w}} \mathrm{CVaR}_{95}(L(\mathbf{w}))$
over KZT/USD/EUR weights
Benchmarks: all KZT, all USD, all EUR, equal weight, income matched, expenditure matched
Output: optimal & near-optimal allocations
simulated FX paths
Robustness analysis
household stress • SB10 / SB56 / IID • 2020–2025 estimation window • monthly / quarterly / no rebalancing • transaction costs: 0%, 0.25%, 0.50%, 1%